

Pragyansh Saxena

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Summary

Software Engineer with 2+ years of experience building production web, mobile, backend, and AI/ML systems across startups, research, and freelance engagements. Delivered systems serving 10,000+ users with 99.9% uptime, reduced API latency by up to 40%, built RAG pipelines over 50,000+ embeddings, and led deep learning research achieving state-of-the-art cross-domain generalization on speaker detection benchmarks. Strong background in backend architecture, mobile development, distributed systems, computer vision, and full-stack engineering using Python, Go, TypeScript, React Native, PyTorch, FastAPI, PostgreSQL, Docker, AWS, and GCP.

Work Experience

Software Engineer

Jun 2024 – Apr 2026

Instafarms — Remote

- Architected and shipped 3 production web and mobile applications using Next.js, React Native, and TypeScript with REST APIs and real-time data sync for 500+ daily active users, reducing API response latency by 35% and onboarding overhead by 20%.
- Designed and deployed horizontally scalable RAG pipelines on GCP with semantic vector search over 50,000+ embeddings, reducing LLM query latency by 40% for 500+ daily active users.
- Provisioned and managed 5 Linux VMs with Docker and Nginx, achieving 99.9% uptime and reducing deployment cycles by 30% through GitHub Actions CI/CD with automated unit and integration test gates.
- Optimized PostgreSQL schemas, composite indexes, and SQL queries, reducing dashboard query execution time by 50% using telemetry-driven performance monitoring.

Full Stack Developer Intern

Mar 2024 – Jun 2024

Dynish — Remote

- Improved frontend performance by 25% through React.js refactoring, Next.js server-side rendering, and client-side optimization, contributing to an 18% increase in user retention.
- Replaced 4 manual data-processing workflows with NLP-powered FastAPI services, saving 15+ engineering hours per week.
- Migrated the authentication layer to Firebase, resolving 60% of recurring login failures and reducing cloud infrastructure costs by \$200/month.
- Contributed across React.js, TypeScript, PostgreSQL, and AWS in a 6-person Agile/Scrum team, increasing sprint velocity by 22%.

Education

Indian Institute of Technology, Mandi

Aug 2023 – May 2027

B.Tech in Computer Science Engineering

CGPA: 8.41

Coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Machine Learning, Deep Learning, Computer Vision, Data Science

Projects

ASD Generalization Study – Cross-Domain Active Speaker Detection (Research) | GitHub

PyTorch, Computer

Vision, Transformers, Audio-Visual Learning

- Led a research team to identify that multi-face inter-speaker context is the dominant driver of cross-dataset overfitting in Active Speaker Detection, validated through 13+ controlled model variants and ablations across two benchmark datasets (AVA-ActiveSpeaker, Columbia).
- Designed a Transformer-based temporal modeling fix that cut cross-domain performance drop from -27.87pp to -0.69pp, and built an adaptive face-count-routing inference system (FCAI) achieving 94.11% AVA mAP with 70.96% Columbia F1 simultaneously.
- Won 1st place at the IIT Mandi CS671 Deep Learning Hackathon; findings are being prepared for publication as a research paper with faculty mentor Dr. Jyoti Nigam (Prof. Aditya Nigam).

NetSched – Network-Aware Distributed Task Scheduler | GitHub

Go, C++, Linux, TCP, Docker

- Designed a network-aware distributed task scheduler that selects execution nodes using CPU/memory availability, queue state, network latency, bandwidth, transfer cost, and failure risk.
- Implemented concurrent Go/C++ scheduler and worker runtimes with custom TCP messaging, task queues, resource accounting, heartbeats, acknowledgements, node-state tracking, and automatic orphan-task recovery.
- Implemented FIFO, Round Robin, Least-Loaded, and network-aware scheduling using priority queues, min-heaps, hash maps, resource admission control, and normalized multi-factor placement costs.

WebSync – AI-Powered Uptime Monitoring Platform | Live | GitHub

Next.js, FastAPI, PostgreSQL, LLMs

- Built an AI-powered uptime monitoring platform using Next.js and FastAPI, implementing automated LLM-based anomaly detection across 200+ endpoints and reducing client downtime by 15%.
- Implemented PostgreSQL event storage and real-time alerting workflows, enabling 3x faster incident response compared with manual monitoring.

Technical Skills

Languages: Python, C++, Go, Java, JavaScript, TypeScript, SQL, Bash, Dart

AI / ML: PyTorch, TensorFlow, Computer Vision, Deep Learning, Transformers, LLMs, RAG, LangChain, Hugging Face, Scikit-learn, OpenCV, Vector Databases, Prompt Engineering

Mobile: React Native, Dart, Cross-Platform App Development, REST API Integration

Backend: FastAPI, Django, Node.js, Express.js, REST APIs, GraphQL, WebSockets, gRPC, Microservices, Asynchronous Programming

Frontend: React.js, Next.js, Tailwind CSS, Redux, Zustand, HTML, CSS

Databases: PostgreSQL, MySQL, MongoDB, Redis, Supabase, Firebase Firestore, SQLite

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Linux, Nginx, GitHub Actions, CI/CD, Firebase, Vercel

Testing & Engineering: Unit Testing, Integration Testing, Functional Testing, Test Automation, Pytest, Jest, Code Coverage, Agile/Scrum, System Design, Software Architecture, RBAC, Access Control

Achievements

- Won 1st place at the IIT Mandi CS671 Deep Learning Hackathon for the ASD Generalization Study, now being prepared for research publication with faculty mentor.
- Secured 2nd place at KrackHack Hackathon (IIT Mandi), building and delivering a working AI prototype within 24 hours among 50+ participating teams.
- Led the Programming Club website rebuild at IIT Mandi, improving page-load performance by 40% for 1,000+ students.
- Delivered the Ranneti Annual Fest registration portal, processing 10,000+ registrations with zero downtime.